

Wilson Academy

1. a) If $U = \{1, 2, 3, 4, 5, 6, 9\}$, $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 5, 6\}$
 - i. Find $A \Delta B$ [2]
 - ii. Show in Venn diagram. [1]
2. After allowing 20% discount on the marked price of a machine and then levying 13% VAT, the final price of the machine becomes Rs. 75000/-. Then
 - a. What is the relationship between S.P, VAT and S.P with VAT. [1]
 - b. Find the marked price of machine. [2]
3. a. Identify the series: $3+6+9+12+ \dots$ [1]
4. a. Factorize: x^2-5x+6 [1]
 b. Find H.C.F : x^3-1, x^4+x^2+1 [2]
5. a. Solve: $2x-3y=6$ [2]
 $x+3y=3$
6. From the following information, answer the following questions.

Wages in Rs	450	500	550	600	650	700	750	800
No. of workers	2	8	4	10	6	5	4	1

- a. Find upper and lower quartile. [3]

Opt. Maths

1. a) Write formula to calculate distance between (x_1, y_1) & (x_2, y_2) . [1]
 2. a) If $A = \begin{bmatrix} -8 & 2 & -4 & 6 \\ 6 & 1 & 3 & 0 \end{bmatrix}$, find $a_{11} + a_{14} + a_{12}$. [1]
 3. a) If $f(x) = 3x^2 - 2x + 2$, find $f(2)$. [1]
 4. b) Prove that: $\frac{1 - \sin x}{\cos x} = \frac{\cos x}{1 + \sin x}$ [2]
 5. Find the standard deviation. [4]
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|-----------------|---|----|----|----|----|----|
| Marks | 5 | 10 | 15 | 20 | 25 | 30 |
| no. of students | 2 | 3 | 5 | 6 | 3 | 1 |
6. In a ΔABC , $A(-2, -3)$, $B(-6, 5)$, $C(3, 2)$. Rotate ΔABC about origin through 90° in clockwise direction. Find the images. Plot ΔABC and $\Delta A'B'C'$ both on the same graph. [4]

Science

MCQ [4×1=4]

1. Which group doesn't belong to Kingdom Protista
 - a. The protistan algae
 - b. Slime moulds
 - c. Protozoa
 - d. Archaeobacteria
 2. The center of reflex action is
 - a. Nerve
 - b. Brain
 - c. Spinal Cord
 - d. Gland
 3. Which theory is not the organic theory of evolution?
 - a. Special creation by God
 - b. Darwin's theory of natural selection
 - c. Lamarck's theory of evolution
 - d. Mutation theory
 4. Acceleration of body moving with constant velocity will be
 - a. Positive
 - b. Negative
 - c. Zero
 - d. None of them
- #### Very short question
1. What is the full form of NARC? [1]
 2. What is binomial system of nomenclature? [1]
 3. What is second law of motion? [1]
 4. What is electrovalent bond? Write any two characters of electrovalent compound. [2]
 5. What is NPK fertilizer? Give any two examples. Write any two harmful effects of excess use of such inorganic fertilizers. [2]